

DL Midsem Notes — Section 2

History of AI, Biological vs Artificial Neurons

Not tested in any PYQ in your set. Quick pass — read once, move on.

1 Theory

1.1 AI Springs and Winters (Rough Timeline)

- **1956** – Dartmouth workshop coins "Artificial Intelligence"; early optimism (Spring).
- **Late 1960s–70s** – Perceptron's limitations exposed (Minsky & Papert show it can't do XOR); funding dries up (**First Winter**).
- **1980s** – Expert systems and backpropagation revive interest (Spring).
- **Late 1980s–90s** – Expert systems don't scale, hardware/data insufficient (**Second Winter**).
- **2012 onward** – AlexNet's ImageNet win, cheap GPUs, huge datasets ignite the current deep learning boom (Spring, ongoing).

The pattern each time: a promising idea outruns the compute/data available to back it up, hype collapses, then hardware or algorithmic advances catch up and the field re-emerges stronger.

1.2 Biological Neuron vs Artificial Neuron

A biological neuron receives signals through **dendrites**, integrates them in the **soma** (cell body), and if the combined signal crosses a threshold, fires an **action potential** down the **axon**, which passes the signal on through **synapses** to the next neuron.

Biological	Artificial
Dendrites (receive signals)	Inputs x_i
Synaptic strength	Weights w_i
Soma (integrates signal)	Weighted sum $\sum w_i x_i + b$
Firing threshold	Activation function (thresholding)
Axon (sends signal onward)	Output h , fed to next layer

1.3 Sigmoid — A First Smooth Threshold

Biological firing is roughly all-or-nothing (spikes), but a hard step function has zero gradient almost everywhere, which is useless for learning by gradient descent. The sigmoid $\sigma(z) = \frac{1}{1+e^{-z}}$ was an early choice precisely because it *looks* like a soft threshold (0 for very negative input, 1 for very positive) while staying smooth and differentiable everywhere — a compromise that makes gradient-based learning possible. (Its drawbacks are covered properly in section 8.)

2 Practice Questions

Q1. Name the two AI winters and, in one line each, why they happened.

Q2. Match each biological structure to its artificial-neuron analogue: dendrite, synapse, soma, axon.

Q3 (concept check). Why was a smooth function like sigmoid preferred over a hard step function once networks needed to be trained by gradient descent?

3 Answers

A1. First winter (late 1960s–70s): Minsky and Papert showed single-layer perceptrons can't solve linearly-inseparable problems like XOR, killing funding. Second winter (late 1980s–90s): expert systems proved too brittle and expensive to scale, and available compute/data couldn't support more ambitious models.

A2. Dendrite $\rightarrow$ inputs x_i ; synapse $\rightarrow$ weight w_i ; soma $\rightarrow$ weighted sum + bias; axon $\rightarrow$ output passed to the next layer.

A3. Gradient descent needs a non-zero, informative derivative at (almost) every point to know which direction to update weights. A hard step function has derivative 0 everywhere except an undefined jump at one point, so no gradient signal ever flows back through it. A smooth, monotonic function like sigmoid still behaves like a threshold but has a well-defined, non-zero gradient across a useful range of inputs.